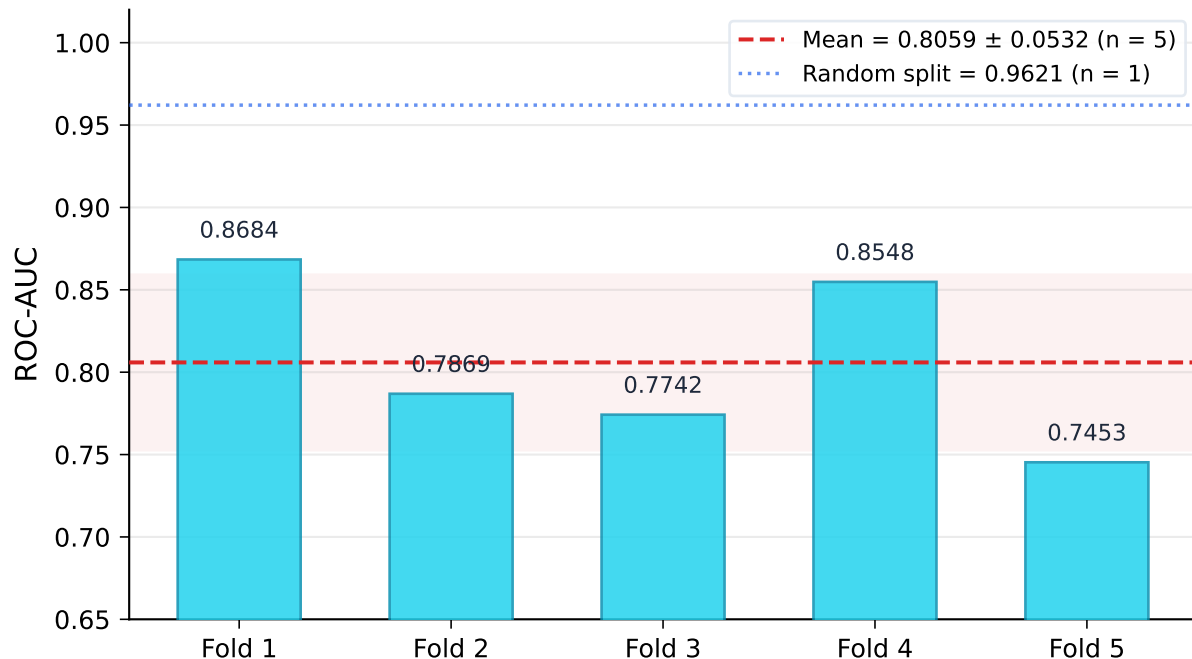


## Clonal Leakage Effect on Model Performance — MIL v5.3 + XGBoost | K. pneumoniae / Ertapenem

### ST-Blocked 5-Fold Cross-Validation



### Performance Loss Under ST-Controlled Evaluation

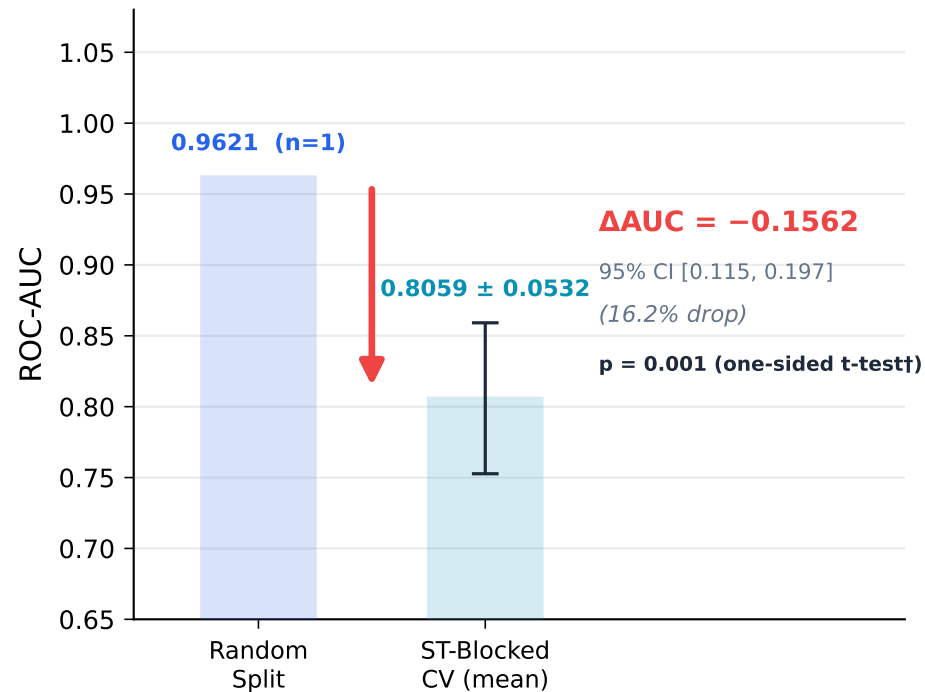


Figure. Random-split evaluation inflates ROC-AUC due to clonal structure in the dataset. Isolates sharing the same sequence type (ST) are distributed across training and test sets under random splitting, allowing the model to exploit clonal similarity rather than learning true resistance determinants. ST-blocked 5-fold CV (mean AUC =  $0.8059 \pm 0.0532$ ) reveals a 16.2% performance drop ( $p = 0.001$ ) relative to the random split, quantifying the magnitude of clonal leakage bias. †One-sided t-test of ST-blocked fold AUCs vs. the random-split point estimate; a paired DeLong test requires per-sample predicted probabilities from both protocols.